Schedule

Assignments:

- Reading assignment Chp 1: Last week
- Reading assignment Chp 3: Today midnight
- Reading assignment Chp 4: Next Monday midnight
- Practice questions: no deadline – no grading

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Schedule

Labs

- Lab 4 on confidence intervals: today with due date 23:59 tomorrow
- Sample case: next Monday in the lab
- Case exam:
  - Monday, November 17, at 16:00-19:00 in labs SNAB149
  - You will be working in groups of 6-7 students. Please form your groups by next Monday (November 10th).
  - Read the information on the case and the sample cases.

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Feedback on Lab 3

Random versus convenience sampling

Target population?

- With or without the outliers?

Agenda

Confidence intervals

- Mean of normal random variables
- Mean of random variables having a general distribution with a large sample size

Statistical Intervals Based on a Single Sample (Chapter 5)

In many situations, a point estimate does not provide enough information about the parameter of interest.

An interval estimate can be more informative regarding to the precision and reliability of estimation.

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Example: IQ test scores and classification

Normal is 100.

Assume that you have a sibling, say X.

X got a score of 98.

You got a score of 101.

Here are some questions:

- Are you the smart guy with superior intelligence?
- Is X dull?

Variability?

IQ Classification
Genius
Very superior intelligence
Superior intelligence
Normal intelligence
Dullness
Border-line deficiency
Definite feeble-mindedness

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Example: IQ test

Average is 100.

Assume that you have a sibling, say X.

X got a score of 98.

You got a score of 101.

Here are some questions:

- Are you the smart guy?
- Is X backward or dull?

Stanford–Binet Fifth Edition (SB5) classification	
IQ Range ("deviation IQ")	IQ Classification
145–160	Very gifted or highly advanced
130–144	Gifted or very advanced
120–129	Superior
110–119	High average
90–109	Average
80–89	Low average
70–79	Borderline impaired or delayed
55–69	Mildly impaired or delayed
40–54	Moderately impaired or delayed

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Confidence Intervals…

In many situations, a point estimate does not provide enough information about the parameter of interest.

An interval estimate can be more informative regarding to the precision and reliability of estimation.

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Remember:

The point estimates for p, where p is the probability of non-defective.



Now the question is:
Can we find an interval that will include the true value of p with some probability?

Normal random sample

$X_1, \ldots, X_n$: random sample

$X_1, \ldots, X_n$ ~ Normal with mean μ and standard deviation σ.

Assume that σ is known.

You want to find an interval that includes the mean μ with probability 1- α.

An interval estimator for μ for $X_i \sim N(\mu, \sigma)$

Our goal is to find L and U such that:

$P(L < \mu < U) = 1 - \alpha$, where L, U are random variables.

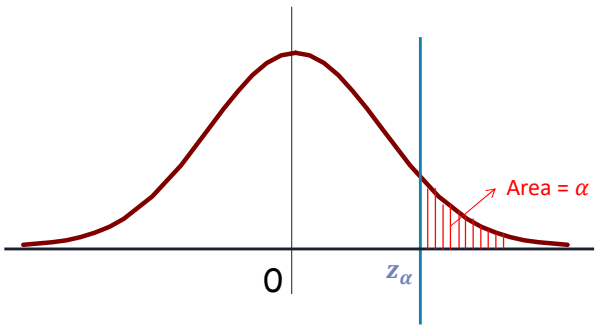
What do we know?

1. An estimator for μ: $\bar{X}$
2. Distribution of the sample mean: $\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

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Definition

z_α : the upper 100α percentage point of the standard normal distribution

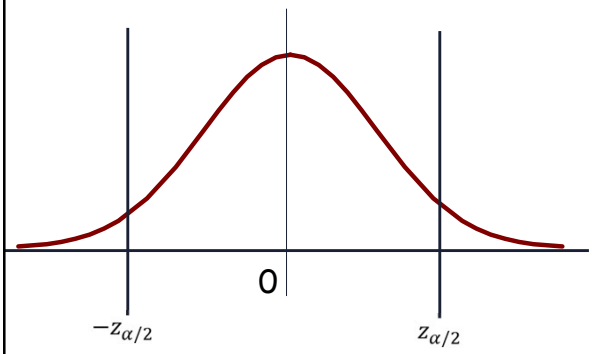


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Question:

$Z \sim N(0,1)$

$P(-z_{\alpha/2} < Z < z_{\alpha/2})$=?



Z in terms of $\bar{X}$?

$\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

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An interval estimator for μ for X_i ~N(μ, σ)

$$P(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}) = 1 - \alpha.$$

L U

probability

In practice, only one random sample is used to construct one confidence interval.

$$P(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}) = 1 - \alpha.$$

$L=l$ $U=u$

confidence

All quantities in this expression are constants, no random variable is involved.

→ NOT REASONABLE TO ATTACH A PROBABILTY TO THIS SPECIFIC EVENT.

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Definition

A 100(1- α)% CI for the mean μ of a normal population when σ is known is given by:

$$\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

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Example: A cargo company

A cargo company aims to have a mean delivery time of 5 hours for local deliveries.

They believe that delivery times are normal with σ=2.4.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

95%CI on the true mean delivery time?

$$\bar{x} = 6.05$$

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The correct interpretation is by many samples via frequency…

Assume that M samples are collected, each having n observations in it.

We use the above formula to calculate a confidence interval for each of these samples.

Here is a description, and
you will do this in the lab yourself…

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M samples, each with n observations, and 1-α as the confidence level

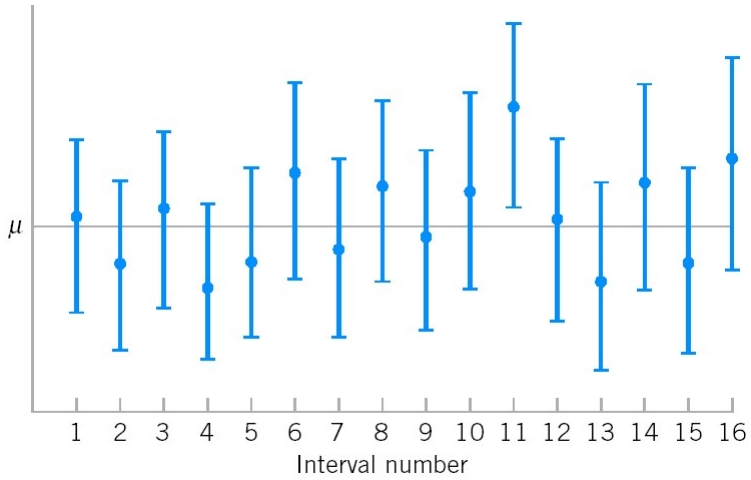
	Observations				Mean	lower bnd	upper bnd
Sample	1	2	…	n	$\bar{x}$	l	u
1	x_1^1	x_2^1	…	x_n^1	$\bar{x}^1$	l^1	u^1
2	x_1^2	x_2^2	…	x_n^2	$\bar{x}^2$	l^2	u^2
M-1	x_1^{M-1}	x_2^{M-1}	…	x_n^{M-1}	$\bar{x}^{M-1}$	l^{M-1}	u^{M-1}
M	x_1^M	x_2^M	…	x_n^M	$\bar{x}^M$	l^M	u^M

How many intervals will cover the real μ?

$\bar{x}^M - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ $\quad$ $\bar{x}^M + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

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Confidence Interval on the Mean of a Normal Distribution, Variance Known



Repeated construction of a confidence interval for μ.

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Confidence Interval on the Mean of a Normal Distribution, Variance Known

Interpreting a Confidence Interval

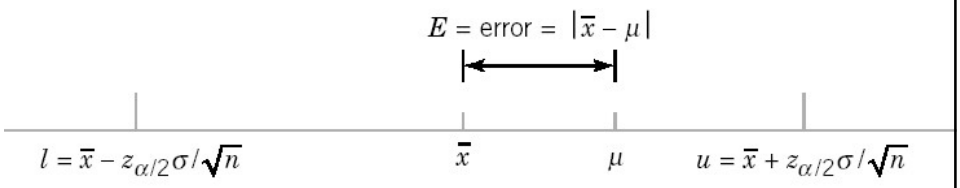
- The confidence interval is a random interval.
- The appropriate interpretation of a confidence interval (for example on μ) is: The observed interval [l, u] brackets the true value of μ, with confidence 100(1-α).

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Confidence Interval on the Mean of a Normal Distribution, Variance Known

Confidence Level and Precision of Error

The length of a confidence interval is a measure of the precision of estimation.



Error in estimating μ with $\bar{x}$.

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$$100(1-\alpha)\%\text{CI:} \quad \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

Confidence Level & Precision

Confidence Level	Width of the Interval
100(1- α)%	$2z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
95%	$2(1.96)\frac{\sigma}{\sqrt{n}}$
99%	$2(2.58)\frac{\sigma}{\sqrt{n}}$
100%	∞

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Confidence Level, Precision & Choice of Sample Size

For fixed sample size: If confidence level is increased, the width of CI is also increased, so that the precision or accuracy decreases if the reliability of CI is increased.

To have both high confidence & high precision, the only way is increasing the sample size.

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Precision & Choice of Sample Size

What is the width of 100(1- α)% CI? $2z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

Let w be the desired width (so the precision you aim) of CI.

What is the required n? $n \geq \left(2z_{\alpha/2}\frac{\sigma}{w}\right)^2$

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Example: A cargo company

A cargo company aims to have a mean delivery time of 5 hours for local deliveries.

They believe that delivery times are normal with σ=2.4.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

If the desired width of 95%CI is 2, how many more delivery times should be recorded?

$$\bar{x} = 6.05$$

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To summarize:

$X_1,\ldots,X_n \sim N(\mu, \sigma)$: random sample

Assumptions: μ is unknown and σ is known.

$$100(1-\alpha)\%\text{CI}: \quad \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

- Interpretation?
- Precision, confidence, sample size?

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The same setting:

$X_1,\ldots,X_n \sim N(\mu, \sigma)$: random sample

Assumptions: μ is unknown and σ is known.

Can we define one-sided CIs?

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One-sided CIs

$P(L < \mu) = 1 - \alpha$, where L is a random variable.

means "probability that *L* is less than m is 1-α".

Now assume that an observation is made on the random variable *L* so that *L=l*.

"$P(l < \mu) = 1 - \alpha$",
where l is an observation of the random variable L.

means "with 100(1-α)% confidence level μ is greater than l"

... and α is called the significance level.

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$100(1-\alpha)\%\text{CI}: \quad \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$

One-sided CIs

$X_1,\ldots,X_n \sim N(\mu, \sigma)$: random sample

Assumptions: μ is unknown and σ is known.

Can you find a lower bound on the population mean μ with 1-α confidence?

Can you find an upper bound on the population mean μ with 1-α confidence?

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Confidence Interval for the Mean of a r.v. with a large sample

$X_1,\ldots,X_n$: random sample

Assumptions:

- n is very large. → $\bar{X} \overset{approx}{\sim} N(\mu, \sigma/\sqrt{n})$ by CLT.
- σ is known.

CI is the same as in normal r.v.s: $\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$

But σ is almost never known!!!

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Confidence Interval for the Mean of a r.v. with Unknown Variance:

$X_1,\ldots,X_n$: random sample (no specification on their distribution)

Assumptions:

- n is very large. → $\bar{X} \overset{approx}{\sim} N(\mu, \sigma/\sqrt{n})$ by CLT.
- both μ and σ are unknown.

↓

σ will be estimated by sample standard deviation S.

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The change in C. I.?

$\frac{\bar{X}-\mu}{S/\sqrt{n}}$ → more variable compared to $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$

but still OK, since n is large!!!

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Proposition

If n is sufficiently large:

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \overset{approx.}{\sim} N(0,1)$$

so that

$$\left(\bar{x}-z_{\alpha/2}\frac{s}{\sqrt{n}}, \bar{x}+z_{\alpha/2}\frac{s}{\sqrt{n}}\right)$$

is a large-sample interval for μ with confidence level approximately $100(1-\alpha)\%$. This formula is valid regardless of the population distribution.

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Rule of thumb

n>40 is generally sufficient for all kinds of random variables. To "fine-tune" this rule for specific probability distributions, you can run simulations.

More conservative than CLT because of the additional variability introduced by using S instead of σ.

You can experiment with "n>40."

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